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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2025

ME1654-1 – INNOVATION AND DESIGN THINKING
(For ME, EC, AC, VL, IS, AM, CC)

Duration: 2 Hours

Max. Marks: 50

*Note: Answer **Three full** questions choosing **One full** question from each **Unit**.*

		Marks	BT*	CO*	PO*
Unit – I					
1.	a) What is design thinking? Explain with an example.	07	L*1	1	2
	b) Explain 9 qualities of a good design.	08	L2	1	2
2.	a) What are the characteristics of a good design? Explain with an example.	07	L1	1	2
	b) Empathetically collect data from a regular college going person looking for a shoe for daily use using 360° approach which can be also used for jogging and playing football.	08	L1	1	2
Unit – II					
3.	a) With a neat sketch show agile model of project management.	07	L1	2	1
	b) With the help of a flow chart explain the process modelling in design thinking.	08	L1	2	1
4.	a) With a neat sketch show waterfall model.	07	L1	2	1
	b) What are the advantages of business process model in design thinking?	08	L1	2	1
Unit – III					
5.	a) Johnson is a bank manager, and he is looking for a walk in wardrobe design for his newly built house. Formals, casuals, semi formals, daily use clothes and other related accessories compartments are must. He has a son who is currently enrolled in college and a daughter who is 4 years old and goes to preschool. His wife maintains a successful career in her corporate field. His mother is a homemaker who has a strong affinity for gold and silver ornaments. His father is a ex-service man retired as a major. Ideate a walk in wardrobe for this family located in one common space so that all of their clothing and other necessary requirements are taken care in one convenient location. Take into account all essential inputs.	20	L3	3	3
6.	There is a 1 acre farm land which is equivalent to 22 numbers of 30X40 Sq ft. site. This land is very close to forest and gets all the serenity and tranquility of nearby forest with some wild animals habitat like forest giant squirrel, monkeys, deer, wild boar, snakes and some varieties of native birds. The land is placed in a downhill making the rain water flow downwards and pass through the channel located at the center of the land. A corporate working engineer is looking to build a farm house on this land. He has plans of having medicinal plants in one section of the land, some fruit garden in the other section, a small lake to cultivate some fishes in the third section which can in turn serve as the source of water for the land, fourth and fifth sections for dog kennels having 3 different breeds and a cow shed holding around 5 cows, respectively. Empathize with the engineer, define his requirements and ideate the farm land layout for his ease of operation and maintenance with necessary assumptions. He is ready to invest not more than 20 lakhs on this land. Also show the respective price allotment to each section.	20	L3	3	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
